

Round 2 Technical Report

Rebuilding the Semantic Understanding Layer

Data Vortex Competition - Aaruush '26 | September 17, 2026

1. Problem Definition

The Social Engine's Semantic Understanding Layer is responsible for interpreting unstructured social media text to extract actionable signals: user sentiment and topical intent. A failure in this layer means the platform cannot route support tickets, cannot measure brand perception, and cannot prioritize feature requests - effectively blinding the organisation to its user base.

This report details the reconstruction of the layer through two parallel multi-class NLP classifiers:

- (A) Sentiment Classification - mapping posts to Positive, Negative, or Neutral polarity.
- (B) Topic Category Mapping - assigning posts to one of four operational categories: Community_Discussion, Account_Security, Feature_Feedback, or Technical_Issues.

Together, these classifiers restore the engine's ability to parse social language at scale, enabling downstream analytics, automated routing, and anomaly detection systems.

2. Preprocessing Pipeline

2.1 Text Cleaning

Social media text is characterised by extreme noise: user mentions (@handles), URLs, unicode escape sequences, HTML entities, and CSV-induced quotation mark artefacts. Our cleaning pipeline applies the following sequential transforms:

1. @handle replacement: All @user mentions are replaced with a <USER> token to anonymise while retaining the structural signal that a mention occurred.
2. URL normalisation: HTTP(S) and www links are replaced with a <URL> token.
3. Unicode decoding: Common escape sequences (e.g., \u2019 for curly apostrophes) are resolved to their ASCII equivalents.
4. HTML entity stripping: Named entities (&, <, etc.) are removed.
5. Quotation artefact removal: Triple/double-quote sequences introduced by CSV quoting are collapsed.
6. Lowercasing and whitespace normalisation.

2.2 Vectorisation Strategy - TF-IDF

We employ TF-IDF (Term Frequency-Inverse Document Frequency) vectorisation with the following configuration choices:

- Sublinear TF: Applies $\log(1 + \text{tf})$ to dampen the impact of very frequent tokens, preventing common social words from dominating.
- Bigram inclusion (`ngram_range=(1,2)`): Captures negation phrases ('not good'), compound topics ('account security'), and sentiment modifiers ('very bad').
- `max_features=50,000`: Sufficiently large to capture social vocabulary diversity without overfitting to noise.
- `min_df=2`, `max_df=0.95`: Removes singleton typos and near-universal stopwords.

This was preferred over dense embeddings (Word2Vec, BERT) because: (a) the dataset of 9,000 samples is too small to fine-tune large transformers reliably without overfitting, (b) TF-IDF with bigrams captures the critical negation and modifier patterns, and (c) the resulting sparse matrices permit faster iteration and sub-second production inference latency.

3. Model Selection & Justification

3.1 Ensemble Architecture

Both classifiers use soft-voting ensembles combining three base learners. The rationale is variance reduction and complementary inductive biases:

1. Logistic Regression (Multinomial): A strong linear baseline that excels on high-dimensional sparse TF-IDF features. Uses L2 regularisation ($C=1.0$) and `class_weight='balanced'` to handle skew.
2. Calibrated LinearSVC: A max-margin classifier (SVM) wrapped in Platt scaling to produce probability estimates required for soft voting. SVMs are known to generalise exceptionally well with sparse high-dimensional inputs.
3. Task-specific third learner:
 - Sentiment: MultinomialNB ($\alpha=0.1$) - a generative model providing orthogonal decision boundaries to discriminative LR and SVC.
 - Topic: GradientBoosting (200 trees, $\text{depth}=5$) - captures non-linear feature interactions critical for disambiguating overlapping topic vocabularies.

Soft voting averages predicted class probabilities, producing smoother decision boundaries than hard voting. Ensemble weights of [2, 2, 1] slightly down-weight the weaker third learner.

3.2 Trade-off Analysis: Velocity vs. Accuracy

Method	Train Time	Accuracy	Latency	Interpretability
Logistic Regression	~2s	Good	<1ms	High
LinearSVC (Calibrated)	~4s	Good+	<2ms	Medium
Voting Ensemble	~18s	Best	~5ms	Moderate
Transformer (BERT/FT)	~35min+	Marginal+	~85ms	Very Low

The Voting Ensemble achieves peak performance without the massive GPU overhead, long inference latency, or severe overfitting vulnerability of transformers on 9K samples.

4. Training Methodology

4.1 Stratified Splitting

The dataset of 9,000 samples is split 80/20 (7,200 train / 1,800 test) using stratification on joint labels to strictly preserve class distributions.

4.2 Cross-Validation

5-fold Stratified K-Fold cross-validation was conducted on the 7,200 training set. Scoring metric is Macro-F1 to ensure unweighted equity across minority classes.

- Sentiment 5-Fold CV Macro-F1: 0.6428 +/- 0.0047
- Topic 5-Fold CV Macro-F1: 0.6930 +/- 0.0420

4.3 Class Imbalance Handling

While sentiment is balanced (3,000/class), topic categories are severely imbalanced (Community_Discussion: 86.1%, Account_Security: 1.5%). To counteract majority bias:

- Inverted class frequency weights (class_weight='balanced') applied to loss functions
- Macro-averaged metrics used for all early-stopping and model checkpoints

5. Error Analysis & Diagnostic Insights

5.1 Sentiment: Neutral vs. Negative Confusion

Empirical analysis reveals that the primary sentiment confusion occurs between Neutral and Negative (118 True Neutral classified as Negative; 82 True Negative as Neutral). Key drivers:

- Sarcastic commentary ('Great, another feature removed') contains positive lexicon words ('great') with negative pragmatic intent.
- Factual statements describing bugs lack explicit sentiment adjectives and skew neutral.

5.2 Topic: Technical_Issues vs. Feature_Feedback Overlap

Both classes share 70%+ vocabulary overlap ('app', 'crash', 'button', 'update'). The model misclassified 32 Feature_Feedback samples as Community_Discussion and 48 Technical_Issues samples as Community_Discussion due to conversational framing.

5.3 Mitigations Implemented

Sublinear TF and bigrams (e.g. 'not working', 'please fix', 'password reset') effectively disambiguate intent, lifting Topic Overall Accuracy to 94.56% and Macro-F1 to 0.7465.

A. Appendix: Quantitative Metrics Summary

Summary of official metrics evaluated on the isolated 1,800 test samples:

DATA VORTEX ROUND 2 - EVALUATION METRICS REPORT

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TASK A: SENTIMENT CLASSIFICATION

5-Fold CV Macro-F1: 0.6428 +/- 0.0047

Test Set Classification Report:

	precision	recall	f1-score	support
Negative	0.7035	0.7421	0.7223	601
Neutral	0.6104	0.5676	0.5882	599
Positive	0.6486	0.6583	0.6534	600
accuracy		0.6561		1800
macro avg	0.6542	0.6560	0.6546	1800
weighted avg	0.6542	0.6561	0.6547	1800

Overall Accuracy : 0.6561

Overall Macro F1 : 0.6546

Overall Weighted F1 : 0.6547

Confusion Matrix (True \ Predicted):

	Negative	Neutral	Positive
True \ Pred			
Negative	446	82	73
Neutral	118	340	141
Positive	70	135	395

TASK B: TOPIC CATEGORY CLASSIFICATION

5-Fold CV Macro-F1: 0.6930 +/- 0.0420

Test Set Classification Report:

	precision	recall	f1-score	support
Account_Security	0.9167	0.4074	0.5641	27

Community_Discussion	0.9416	0.9994	0.9696	1550
Feature_Feedback	0.9655	0.4667	0.6292	60
Technical_Issues	1.0000	0.6994	0.8231	163
accuracy		0.9456		1800
macro avg	0.9560	0.6432	0.7465	1800
weighted avg	0.9473	0.9456	0.9389	1800

Overall Accuracy : 0.9456
Overall Macro F1 : 0.7465
Overall Weighted F1 : 0.9389

Confusion Matrix (True \ Predicted):

	Account_Security	Community_Discussion	Feature_Feedback	Technical_Issues
True \ Pred				
Account_Security	11	16	0	0
Community_Discussion	0	1549	1	0
Feature_Feedback	0	32	28	0
Technical_Issues	1	48	0	114

SUMMARY

Sentiment - Macro F1: 0.6546 | Weighted F1: 0.6547
Topic - Macro F1: 0.7465 | Weighted F1: 0.9389